Retail Orders: Queries by Hand (Worked Answers)

Fall 26

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How to work out each answer by hand from the three tables.

customers

customer_id	first_name	last_name
1	John	Doe
2	Jane	Smith
3	Robert	Johnson
4	Sarah	Williams

products

product_id	name	price
1	Silk Blouse	89.99
2	Denim Jeans	79.99
3	Cashmere Sweater	149.99
4	Cotton T-Shirt	29.99
5	Wool Coat	199.99

orders

order_id	customer_id	product_id	order_date
1	1	3	2025-03-10
2	2	1	2025-03-12
3	3	4	2025-03-15
4	2	4	2025-03-18
5	1	2	2025-03-20
6	2	3	2025-03-25

Question 1

Show the `first_name` and `last_name` of every customer who placed an order between 2025-03-15 and 2025-03-21. List each customer only once.

1. Go through the `orders` table and find the orders with an `order_date` from 2025-03-15 to 2025-03-21. Both end dates count.

order_id	customer_id	order_date
3	3	2025-03-15
4	2	2025-03-18
5	1	2025-03-20

2. Write down the `customer_id` of each of those orders: 3, 2, and 1. If a `customer_id` appeared twice, you would write it only once.
3. Look up each `customer_id` in the `customers` table to get the `first_name` and `last_name`.

Answer

first_name	last_name
Robert	Johnson
Jane	Smith
John	Doe

Question 2

For every order of a product that costs more than $80, show the customer's `first_name`, the product's `name`, and the `order_date`.

1. In the `products` table, find the products that cost more than $80: Silk Blouse (`product_id` 1), Cashmere Sweater (`product_id` 3), and Wool Coat (`product_id` 5).
2. In the `orders` table, find every order whose `product_id` is 1, 3, or 5.

order_id	customer_id	product_id	order_date
1	1	3	2025-03-10
2	2	1	2025-03-12
6	2	3	2025-03-25

   No one ordered the Wool Coat, so it does not appear.

3. For each of those orders, look up the `customer_id` in `customers` to get the `first_name`, and the `product_id` in `products` to get the `name`. Keep the `order_date` from the order.

Answer

first_name	name	order_date
John	Cashmere Sweater	2025-03-10
Jane	Silk Blouse	2025-03-12
Jane	Cashmere Sweater	2025-03-25

Question 3

For each product that was ordered at least once, show its `name` and `total_revenue`, its price times the number of times it was ordered, sorted from highest `total_revenue` to lowest.

1. Count how many times each `product_id` appears in the `orders` table.

product_id	name	price	times ordered
1	Silk Blouse	89.99	1 (order 2)
2	Denim Jeans	79.99	1 (order 5)
3	Cashmere Sweater	149.99	2 (orders 1 and 6)
4	Cotton T-Shirt	29.99	2 (orders 3 and 4)
5	Wool Coat	199.99	0

2. Leave out the Wool Coat, because it was never ordered.
3. For each remaining product, multiply the `price` by the number of times it was ordered to get its `total_revenue`.
   - Silk Blouse: $89.99 \times 1 = 89.99$
   - Denim Jeans: $79.99 \times 1 = 79.99$
   - Cashmere Sweater: $149.99 \times 2 = 299.98$
   - Cotton T-Shirt: $29.99 \times 2 = 59.98$
4. Put the rows in order from the highest `total_revenue` to the lowest.

Answer

name	total_revenue
Cashmere Sweater	299.98
Silk Blouse	89.99
Denim Jeans	79.99
Cotton T-Shirt	59.98

Question 4

Show the `first_name` and `num_orders` of every customer who placed more than one order.

1. Count how many times each `customer_id` appears in the `orders` table.

customer_id	first_name	orders
1	John	2 (orders 1 and 5)
2	Jane	3 (orders 2, 4, and 6)
3	Robert	1 (order 3)
4	Sarah	0

2. Keep only the customers with more than one order: John and Jane. Robert has exactly one order, so he is left out.

Answer

first_name	num_orders
John	2
Jane	3

Question 5

Show the `first_name` and `last_name` of every customer who has never ordered the Cotton T-Shirt.

1. In the `products` table, find the `product_id` of the Cotton T-Shirt: 4.
2. In the `orders` table, find every order with `product_id` 4: order 3 (`customer_id` 3) and order 4 (`customer_id` 2). So Robert and Jane bought the T-Shirt.
3. Go through the `customers` table and cross out Robert and Jane. The customers who are left never ordered the T-Shirt.

customer_id	first_name	last_name	Ordered the T-Shirt?
1	John	Doe	no
2	Jane	Smith	yes
3	Robert	Johnson	yes
4	Sarah	Williams	no

4. Sarah has no orders at all, so she never ordered the T-Shirt either. She belongs in the answer.

Answer

first_name	last_name
John	Doe
Sarah	Williams